Comments on the Bill Example

- How many ways are there to select 5 bills from 7 denominations, assuming there are enough bills of each denomination
- **An alternative solution**
  Given that there are 7 choices for the first bill, 7 choices for the second bill,…., 7 choices for the fifth bill, somebody could argue that
- The total number of ways is 7^5.
- What is wrong with this solution?
- Lets take a choice by the alternative solution : e.g., (7,7,1,1,1) means that for the first 3 bills, I select the first denomination, while for the next 2, I select the 7^{th}.
- Any other choice, e.g., (1,7,1,1,7) with the same number of 1s and 7 is considered distinct by the alternative solution.
- But it should not be distinct because the order in which the bills are selected is not important. Bills of the same denomination are indistinguishable.
- The bars and stars representation eliminates this problem.
- All choices with three 1s and two 7s have the same representation and are counted once.
- Representation of (1,7,1,1,7) (7,7,1,1,1) and all equivalent choices using bars and stars:**||||||***